MAULANA HAMIDY CHASH CHASH AL HAQUE

Bandung, Indonesia | (+62) 812-3330-9023 | mhcalhaque@gmail.com | linkedin.com/in/cc-alhaque

Software Engineer | Backend Engineer | AI & Machine Learning

Python | Django | Node.js | PostgreSQL | Computer Vision

Education

Bandung, Indonesia **Institut Teknologi Bandung** **August 2014–June 2020**

- Undergraduate: Aerospace Engineering Major. Cumulative GPA: 3.02.
- Coursework: Mathematics; Introduction to Information Technology; Engineering Mathematics I & II; Statistics; Engineering Analysis and Numerical Method; New Product Development; Computational Aerodynamics; Control Theory; Creativity and Innovation; Industrial Management.
- Thesis : Computational Fluid Dynamics on Wings with Flapping Motion

Yogyakarta, Indonesia **Universitas Gadjah Mada** **September 2022–July 2024**

- Master: Computer Science Major. Cumulative GPA: 3.82.
- Coursework: Analysis of Algorithm; Mathematics for Computer Science; Cryptology; Operation Research; Verification & Validation; Data Warehouse & Business Intelligence; AI Principle; Data Science; Computational Intelligence & Machine Learning; Decision Support System.
- Thesis : Optimizing Adaptive Serial Combination Model Using Genetic Algorithm for Financial Crisis Prediction

Taoyuan, Taiwan **National Central University** **September 2023–July 2024**

- Master Dual Degree: Computer Science Major. Cumulative GPA: 94.4/100.
- Coursework: Introduction to Deep Learning; Data Science Practice.
- Thesis : Adaptive Serial Combination Model Using Genetic Algorithm for Financial Crisis Prediction

Technical Skills

- **Programming: Python, Node.js, Javascript, Typescript, SQL, MATLAB, C++, C#, HTML, CSS**
- **Backend Development: Django, REST API, API Integration, ERP Development, PostgreSQL**
- **Artificial Intelligence & Machine Learning: PyTorch, TensorFlow, Scikit-learn, OpenCV, CNN, LSTM, Autoencoder, Computer Vision, Time-Series Forecasting, Image Segmentation, Feature Engineering, Model Evaluation**
- **Cloud & DevOps: Docker, Git, Azure DevOps, Linux**
- **Tools: Power BI, Odoo, PostgreSQL**

Technical Experience

Professional Experience

- **Software Developer at Stamps** (April 2026 – July 2026)
  - Develop and maintain enterprise ERP/POS backend systems using Python, Django, and PostgreSQL.
  - Design and implement RESTful APIs supporting purchasing, inventory, warehouse, and goods receipt processes.
  - Investigate and resolve production issues through debugging, root cause analysis, and performance optimization.
  - Implement new backend features while ensuring maintainability, scalability, and code quality.
  - Collaborate with Product, QA, and Business teams throughout software development and deployment.
- **Part-time Computer Science Lecturer at Cakrawala University** (February 2026 – July 2026)
  - Teach undergraduate courses in Programming Fundamentals, Programming for Business, Machine Learning, and Differential Calculus for Data Science.
  - Design course syllabi, lecture materials, laboratory exercises, programming assignments, and project-based assessments.
  - Deliver hands-on instruction in Python programming, software engineering fundamentals, machine learning algorithms, and mathematical foundations for data science.
  - Mentor students through practical software development and AI projects while promoting analytical problem-solving and industry best practices.
- **Software Engineer (ML/CV) at Revolution Aerospace** (October 2025 – March 2026)
  - Developed faster-than-realtime flight encounter simulation systems for Detect-and-Avoid (DAA) research.
  - Built API-integrated flight guidance and alerting software for autonomous aviation applications.
  - Integrated tracker uncertainty models into simulation pipelines for more realistic flight scenarios.

- Applied machine learning and computer vision concepts to UAV software development.
- **System Engineer at Tunas Ridean** (January 2025 – July 2025)
  - Developed ERP modules supporting appointment scheduling, workshop operations, sales, and reception.
  - Implemented inventory automation for fast-moving, slow-moving, non-moving, and dead stock classification.
  - Developed OTP verification features supporting Private Data Protection initiatives.
  - Automated report generation and FTP-based distribution through scheduled background processes.
- **Teaching Assistant for Data Science Practice Course at National Central University** (February 2024 – July 2024)
  - Assisted graduate-level Data Science Practice laboratories covering Python, machine learning, and data analysis.
  - Guided students through practical AI assignments and project implementation.
- **Software Engineer at Paperplane.id** (September 2022 – March 2024)
  - Developed educational software supporting curriculum development and Olympiad preparation.
  - Implemented backend features and maintained production systems.
  - Collaborated with software engineers to deliver scalable education technology solutions.
- **Drone System Engineer at SAS Aero Sishan** (August 2020 – January 2022)
  - Developed UAV simulation and flight software supporting autonomous drone operations.
  - Performed system testing and engineering validation for unmanned aerial vehicles.

Projects

- **Enterprise Point of Sale (POS) Platform** (2026–Present). Backend development and maintenance for enterprise Point of Sale modules, implementing RESTful APIs, business logic, production enhancements, and system integrations using Python, Django, and PostgreSQL.
- **Customer Relationship Management (CRM) Platform** (2026–Present). Develop backend features supporting customer management, sales operations, and CRM workflows while maintaining production stability and implementing new business requirements.
- **Enterprise Business Systems** (2026–Present). Develop backend modules supporting purchasing, inventory, warehouse, goods receipt, employee management, attendance, and notification services across enterprise applications.
- **Faster-than-realtime Flight Encounter Simulation** (2025-2026). Developed a high-performance Detect-and-Avoid (DAA) simulation platform with aviation API integration for autonomous flight scenario evaluation.
- **Flight Guidance and Alerting System** (2026). Designed and implemented flight guidance and voice alerting software supporting autonomous aviation applications.
- **Financial Crisis Prediction using Adaptive Serial Combination Model** (2022–2024). Designed a machine learning framework combining Genetic Algorithm optimization and ensemble learning for corporate financial crisis prediction.
- **Workshop ERP System** (Tunas Daihatsu, 2025). Developed enterprise ERP modules supporting workshop operations, appointment scheduling, inventory movement analysis, and reporting automation.
- **Education Technology Platform** (Paperplane, 2022–2024). Developed backend software supporting curriculum management and Olympiad learning platforms.
- **Medical Image Segmentation** (2023). Implemented machine learning techniques for cancer cell image segmentation using computer vision.
- **Cost Estimation Web Application** (Pertamina, 2024). Developed frontend components supporting engineering cost estimation workflows.

Additional Experience and Awards

Awards

- **Silver Medal at Singapore Mathematical Olympiad (2010)**
- **Bronze Medal at OSN Mathematics SMP (2010)**
- **2nd Winner at National Logic Competition (NLC) Institut Teknologi Sepuluh Nopember (2013)**
- **5th Place at Olimpiade Matematika ITS (OMITS) (2013)**
- **5th Place at National Programming and Logic Competition (NPLC) Universitas Surabaya (2013)**
- **Best Design and Best Idea of Fixed Wing Category at Kompetisi Robot Terbang Indonesia (KRTI) (2016)**
- **Technology Complexity Award of Racing Plane Category at Kompetisi Robot Terbang Indonesia (KRTI) (2017)**

Experiences

- **Teaching Assistant Experiences at Courses**

- **(2017) Fluid Mechanics for S1 (Undergraduate) Aerospace Engineering ITB**
- **(2022) Python Language Training for S2 (Master) Computer Science UGM**

- **(2024) Data Science Practice for Master in Computer Science NCU Taiwan**
- **Mathematics Olympiad Teacher (2015 – present)**
- **(2018-2021) Mathematics and Physics Tutor at Bandungtutors**
- **Managerial Experiences**
- **Head of Unmanned Aerial Vehicle (UAV) Division at Keluarga Mahasiswa Teknik Penerbangan (KMPN) ITB (2017-2018)**
- **Field Coordinator (Korlap) at Wisuda Juli KMPN ITB (2017)**
- **Business Owner at Borjuice Bar and Anoman Chips (2021)**
- **(2015-2016) UAV Manufacturer at Rebellion Spy ITB**
- **(2016-2017) UAV Design Engineer at Aksantara ITB**
- **(2019) Student Intern at Dirgantara Indonesia**